

## Use Cases

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**Use Case<sub>1</sub>****Primary Actor****Goal in Context****Preconditions****Trigger**

Initiate Gameplay

Player

To initiate the “play” state of the game where the player gets to control the character.

Program must be booted up and in the main UI state.

The player clicks the “play” button on the main UI screen.

**Scenario:**

- Player: Observes main UI
- Player: Clicks the “play” button
- System: Transition to “play” state

**Exceptions: (N/A)****Priority****When Available****Frequency of Use****Channel to Actor**

Essential, must be implemented

In the main UI state

Once per play session

Via main UI

**Open Issues:**

1. Should the game boot up right away into the “play” state or should there be a state in between main UI and play?
2. Does clicking play start at level one or does it take into account character save state?

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**Use Case<sub>2</sub>****Primary Actor****Goal in Context****Preconditions****Trigger**

Moving the burglar around

Player

To move the burglar around to interact with objects in game using the arrow keys (or WASD).

Player must be in the “play” state.

Player is in the “play” state of the game.

**Scenario:**

- Player: Presses an arrow key
- System: If tile is open then the player moves in the direction of the arrow
- Player: Collects reward, punished by enemy or nothing

### Exceptions:

- The tile the player tries go to is occupied by a border: the player does not move that tile.
- A reward is on the tile the player tries to go to: the given score on the reward is added to the player's score total and the player move to the tile.
- An enemy is on the desired tile: player receives the punishment associated with the enemy and the player moves to the tile.
- Nothing on the desired tile: player moves to the tile.

|                         |                                |
|-------------------------|--------------------------------|
| <b>Priority</b>         | Essential, must be implemented |
| <b>When Available</b>   | During the "play" state        |
| <b>Frequency of Use</b> | Many times, per play session   |
| <b>Channel to Actor</b> | Via the main game loop         |

### Open Issues:

1. Can the player move with any desired keys, maybe can change key binds in settings?
2. Will there be diagonal movement functionality, would probably have to include hitboxes for enemies/player?
3. Display a warning when the player tries to go to an inaccessible location?
4. Does hitting a static enemy remove it from the tile?

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|-----------------------------|-----------------------------------------------------|
| <b>Use Case<sub>3</sub></b> | Completing a level                                  |
| <b>Primary Actor</b>        | Player                                              |
| <b>Goal in Context</b>      | To "win" the level (pull off a successful robbery). |
| <b>Preconditions</b>        | Player must be in the "play" state.                 |
| <b>Trigger</b>              | Player gains a high enough score.                   |

### Scenario:

- Player: Collects a reward to push him over the score threshold
- System: Alert enemies that the score threshold has been reached
- System: Allow player to use the exits
- Player: Finds the nearest exit and escapes without interacting with an enemy
- Player: Advances to next level or wins

### Exceptions:

- Player escapes without being caught by the homeowner:
  - If on last level: display "win" UI.
  - If not on last level: advance to next level.
- Player is caught by homeowner before escaping: restart current level and reset score appropriately.

|                         |                                                      |
|-------------------------|------------------------------------------------------|
| <b>Priority</b>         | Essential, must be implemented                       |
| <b>When Available</b>   | During the "play" state and high enough score        |
| <b>Frequency of Use</b> | Rarely, only once at the end of every level          |
| <b>Channel to Actor</b> | Textbox telling the player that it is time to escape |

**Open Issues:**

1. What if the player wants to continue exploring/stealing high than the required amount to escape?
  2. Is the homeowner alerted that the player has reached the required score?
  3. Will there be one or multiple exits that the player can escape from?
  4. Can players go back to levels they previously escaped?
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**Use Case<sub>4</sub>**  
**Primary Actor**  
**Goal in Context**

Making noise  
Player  
When the player makes noise (interacting with an enemy whose punishment is noise) certain enemies may react to the noise.  
Player must be in the “play” state.  
Player interacts with an enemy that makes noise.

**Preconditions**  
**Trigger**

**Scenario:**

- Player: Moves to a tile with a noise-based enemy
- System: Noise is created and other enemies are attracted to the player’s location

**Exceptions: (N/A)**

**Priority**  
**When Available**  
**Frequency of Use**  
**Channel to Actor**

Important  
During the “play” state  
Multiple times, per play session  
Interacting with a noise enemy

**Open Issues:**

1. Do all enemies make some noise?
  2. Is noise going to be a hidden stat of the player? i.e. the more noise you make the more prone you are to attracting the enemies, and when you stop making noise the stat slower starts to decrease.
  3. Does creating noise randomly attract enemies or is it always going to attract enemies?
  4. Does collecting rewards make noise?
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**Use Case<sub>5</sub>**  
**Primary Actor**  
**Goal in Context**

Altering score  
Player  
To increase or decrease the amount of score the player has.  
Player must be in the “play” state.  
Player interacts with a scoring altering object.

**Preconditions**  
**Trigger**

**Scenario:**

- Player: Moves to a tile with a score boosting/removing object
- System: Calculate and apply score increases/decreases

**Exceptions:**

- The object the player interacts with is negative: player score decreases by set amount determined by object.
- The object the player interacts with is positive: player score increases by set amount determined by object.

|                         |                                     |
|-------------------------|-------------------------------------|
| <b>Priority</b>         | Essential, must be implemented      |
| <b>When Available</b>   | During the “play” state             |
| <b>Frequency of Use</b> | Many times, per play session        |
| <b>Channel to Actor</b> | Interaction with score-based object |

**Open Issues:**

1. Will score increase/decrease be hard coded or have formulaic way of determining it?
2. Can the player have negative score? Does negative score = lose?
3. Is there a max score?

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|-----------------------------|-----------------------------------------------------------------------|
| <b>Use Case<sub>6</sub></b> | Homeowner Interaction                                                 |
| <b>Primary Actor</b>        | Player                                                                |
| <b>Goal in Context</b>      | When the homeowner catches the player must restart the current level. |
| <b>Preconditions</b>        | Player must be in the “play” state.                                   |
| <b>Trigger</b>              | Player interacts/collides with the homeowner object.                  |

**Scenario:**

- Player: Is on the same tile as the homeowner
- System: Enter “game over” state

**Exceptions:**

- The player moves to the tile the homeowner is occupying: restart level.
- The homeowner moves to the tile the player is occupying: restart level.

|                         |                                   |
|-------------------------|-----------------------------------|
| <b>Priority</b>         | Essential, must be implemented    |
| <b>When Available</b>   | During the “play” state           |
| <b>Frequency of Use</b> | 0-Multiple, per play session      |
| <b>Channel to Actor</b> | Interaction with homeowner object |

**Open Issues:**

1. Does the homeowner path find to the player?
2. Are there multiple homeowners on any given level?

### 3. Does noise affect the homeowner?

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|-----------------------------|------------------------------------------------------------|
| <b>Use Case<sub>7</sub></b> | Game over via negative score                               |
| <b>Primary Actor</b>        | Player                                                     |
| <b>Goal in Context</b>      | Game ends without homeowner interaction                    |
| <b>Preconditions</b>        | Player must be in the “play” state: current score is low.  |
| <b>Trigger</b>              | Player interacts/collides with an object that lowers score |

#### Scenario:

- Player: Moves onto a tile containing a score reducing object
- System: Check if current score is now below zero
- System: Move into “game over” state
- System: Displays game over message

#### Exceptions:

- Score hits exactly 0: The game continues

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|-------------------------|--------------------------------|
| <b>Priority</b>         | Essential, must be implemented |
| <b>When Available</b>   | During the “play” state        |
| <b>Frequency of Use</b> | 0-few, per play session        |
| <b>Channel to Actor</b> | “Game over” state              |

#### Open Issues:

1. Warning sound when score is low?
  2. Pulse mechanic when score is low?
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|-----------------------------|-------------------------------------------------|
| <b>Use Case<sub>8</sub></b> | Pausing the game                                |
| <b>Primary Actor</b>        | Player                                          |
| <b>Goal in Context</b>      | To temporarily halt the in-game time            |
| <b>Preconditions</b>        | Player must be in the “play” state              |
| <b>Trigger</b>              | Player presses “escape” or “P” key on keyboard. |

#### Scenario:

- Player: Presses pause key
- System: Game loop is frozen (all ticks stop)
- System: Pause menu is displayed
- System: Game is in “paused” state

#### Exceptions: (N/A)

|                         |                         |
|-------------------------|-------------------------|
| <b>Priority</b>         | Medium                  |
| <b>When Available</b>   | During the “play” state |
| <b>Frequency of Use</b> | 0-few, per play session |

**Channel to Actor**

Pause UI and keyboard input

**Open Issues:**

1. Does pausing take you to a whole new or just freeze the game?